Part Three

QUANTUM SPACE AND RELATIONAL TIME

If you have followed me this far, you now have all the elements with which to understand the current image of the world suggested by fundamental physics – its power, its weaknesses, its limits.

There is a curved spacetime born 14 billion years ago – nobody knows how – and still expanding. This space is a real object, a physical field with its dynamics described by Einstein's equations. Space bends and curves under the weight of matter and plunges into black holes when matter is too concentrated.

Matter is distributed in 100 billion galaxies, each containing 100 billion stars, and is made up of quantum fields which manifest themselves in the form of particles, such as electrons and photons, or as waves, such as the electromagnetic ones that bring us television images and the light of the Sun and the stars.

These quantum fields make up atoms, light and the full contents of the universe. They are strange objects: their quanta are particles that appear when they interact with something else; left alone, they unfurl into a 'cloud of probability'. The world is a swarming of elementary events, immersed in the sea of a vast dynamical space which sways like the water of an ocean.

With this image of the world, and the few equations that make it concrete, we can describe almost everything that we see.

Almost. Something is missing. And it is this something that we are seeking. The rest of the book talks about this missing part.

Turning the page, you pass from what, for good or ill, we credibly know about the world, to what we don't yet know but are trying to glimpse.

Turning the page is like leaving the security of our small spacecraft of near-certainties and stepping into the unknown.